CLIP: Caption–Image Similarity and Zero-Shot CIFAR-10 Classification

DATA 266 · Ritika Mukesh Neema · San José State University · November 13, 2026

Summary. CLIP ViT-B/32 ranks dog images first for the caption “a photo of a dog” (ROC-AUC 0.963, Precision@100 0.76). Used as a zero-shot classifier with an 8-prompt ensemble, it reaches **89.31% top-1** and 97.67% top-3 accuracy on the 10,000-image CIFAR-10 test set without any training. Most errors are between look-alike animals, led by deer→horse (13.1% of deer).

1. Setup

Setting	Value
Model	openai/clip-vit-base-patch32 (151.3M params, 512-d shared embedding, logit scale 100)
Dataset	CIFAR-10 test split: 10,000 images, 32×32 RGB, 10 classes × 1,000
Environment	Google Colab, NVIDIA A100 40GB, PyTorch 2.14.1, transformers 5.22.1, fp16 inference
Seed	6638
Runtime	341 s end to end; encoding 1,000 images 4.6 s, 10,000 images 7.7 s

2. Part 1 — Caption vs. 1,000 images

Method

1. **Load the CLIP model** and processor from Hugging Face (fp16 on GPU).
2. **Define the caption:** “a photo of a dog” (target class: dog).
3. **Preprocess multiple images:** 100 images per class sampled from the test split with seed 6638 (1,000 total). The processor resizes each 32×32 image to 224×224 (bicubic), center-crops and normalizes with CLIP's channel mean/std.
4. **Extract and normalize embeddings:** image embedding = visual projection of the vision encoder's pooled output; text embedding = text projection of the text encoder's pooled output. Both are L2-normalized (norms = 1.0000).
5. **Compute similarity:** cosine similarity = dot product of unit vectors; CLIP's scaled logit = 100 × cosine. Retrieval quality is measured by Precision@100 (share of dogs among the 100 highest-scoring images) and ROC-AUC (dog vs. rest).
6. **Visualize** the ranked images with their scores, the per-class score distribution, and (extension) a 10×10 matrix scoring the same images against one caption per class.

Results

Class	Mean cosine	Mean scaled logit	Rank
dog (caption class)	0.2611	26.11	1
cat	0.2401	24.01	2
deer	0.2347	23.47	3
frog	0.2292	22.92	4
bird	0.2269	22.69	5
horse	0.2191	21.91	6
airplane	0.2027	20.27	7
automobile	0.1978	19.78	8
truck	0.1949	19.49	9
ship	0.1933	19.33	10

Table 1. Mean similarity of each class's 100 images to “a photo of a dog”.

Metric	Value
Cosine range, all 1,000 images	0.155 – 0.294 (mean 0.220)
Margin: dog − closest class (cat)	+0.021
Precision@100	0.76
ROC-AUC (dog vs. rest)	0.9626
Non-dog images in top 100	cat 14, deer 8, bird 2 (no vehicles)
Extension: one caption per class, argmax accuracy	87.9% (matching caption highest for 10/10 classes)

Table 2. Retrieval metrics for the caption.



Figure 1. The 18 images most similar to “a photo of a dog”, with cosine scores. Orange frames mark true dogs; the top image is a deer and a cat ranks 4th.

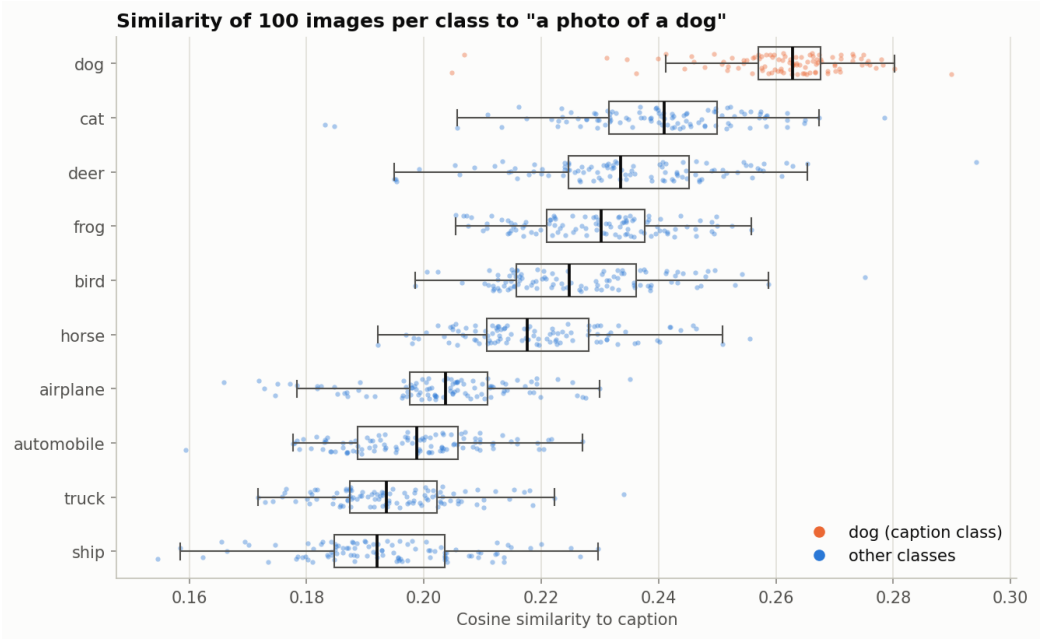


Figure 2. Cosine similarity of all 100 images per class to the caption. Dog is shifted right but overlaps cat and deer.

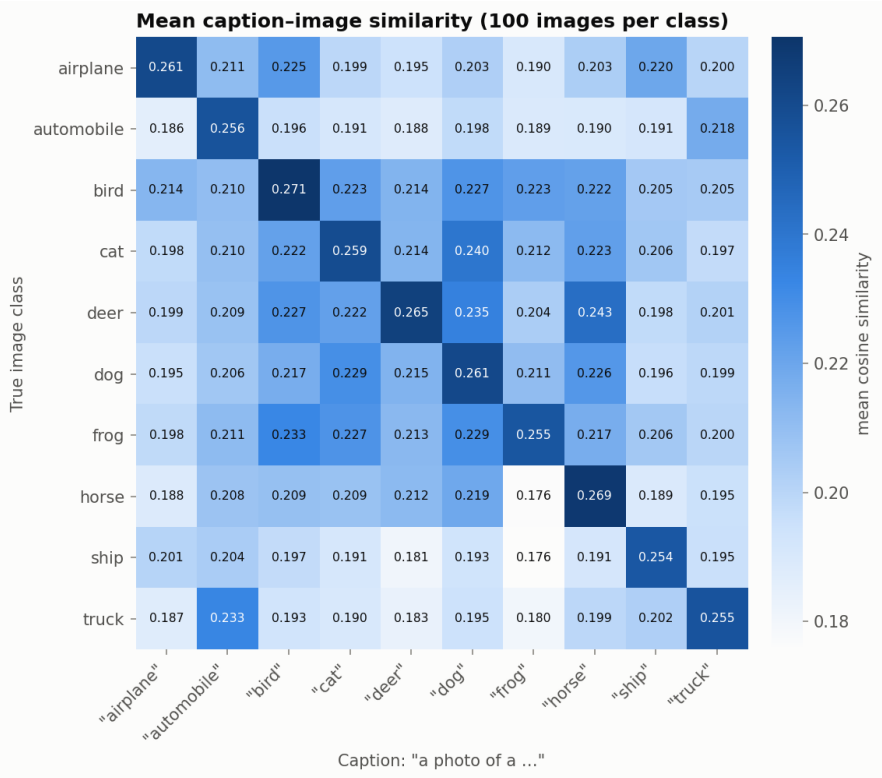


Figure 3. Extension: mean similarity of each image class (rows) to each class caption (columns). The diagonal is the row maximum for all 10 classes.

Findings

1 **Raw cosine scores are compressed; ranking matters.** All scores fall in 0.155–0.294, and dog leads cat by only 0.021. CLIP's logit scale of 100 turns that into a 2.1-logit gap, about 8× in probability, so small cosine differences carry real signal.

2 **Animals cluster, vehicles trail.** After dog come cat, deer, frog, bird and horse; all four vehicle classes score lowest (0.193–0.203). The embedding space separates animals from vehicles although CLIP never saw CIFAR-10 labels.

3 **Strong ranking, overlapping distributions.** All 24 non-dog images in the top 100 are animals. The top-scoring image is a deer (0.294), a small brown four-legged animal on a plain background. A fixed cosine threshold would misclassify many images, but relative ranking works well (ROC-AUC 0.963).

4 **Low resolution limits detail.** Upsampling 32×32 images to 224×224 removes the cues (ears, snout, fur texture) that separate dogs from cats or deer, which is why the false positives are look-alike animals.

5 **One caption per class gives a classifier.** Picking the best of ten class captions labels the 1,000-image subset with 87.9% accuracy. Part 2 evaluates this on the full test set.

3. Part 2 — Zero-shot classification on CIFAR-10

Method

Each class gets a text classifier weight: the class name is inserted into prompt templates, the normalized text embeddings are averaged across templates (prompt ensembling) and re-normalized. Each of the 10,000 test images is assigned the class with the highest cosine similarity; probabilities are the softmax of 100 × cosine. No CIFAR-10 training images or labels are used. A single template, “a photo of a {class}.”, is compared with an 8-template ensemble:

a photo of a {} · a blurry photo of a {} · a low resolution photo of a {} · a pixelated photo of a {} · a photo of the small {} · a photo of the big {} · a good photo of a {} · a close-up photo of a {}

Metrics: top-1 and top-3 accuracy, per-class accuracy (recall), precision, and the confusion matrix.

Results

Prompting	Top-1 accuracy	Top-3 accuracy	Mean per-class accuracy
Single template	88.31%	97.46%	88.31%
8-template ensemble	**89.31%**	97.67%	89.31%

Table 3. Overall zero-shot accuracy on the 10,000 test images.

Class	Accuracy, single	Accuracy, ensemble	Change (pts)	Precision, ensemble
horse	97.1%	**97.6%**	+0.5	83.4%
ship	95.9%	**96.4%**	+0.5	93.1%
automobile	93.3%	**94.4%**	+1.1	92.5%
truck	94.9%	**94.1%**	-0.8	93.9%
airplane	88.9%	**90.7%**	+1.8	95.3%
dog	89.1%	**88.4%**	-0.7	84.3%
bird	89.7%	**88.2%**	-1.5	84.6%
cat	85.7%	**86.5%**	+0.8	81.5%
deer	77.0%	**80.1%**	+3.1	91.0%
frog	71.5%	**76.7%**	+5.2	97.6%

Table 4. Per-class accuracy (recall, 1,000 images per class), sorted by ensemble accuracy.

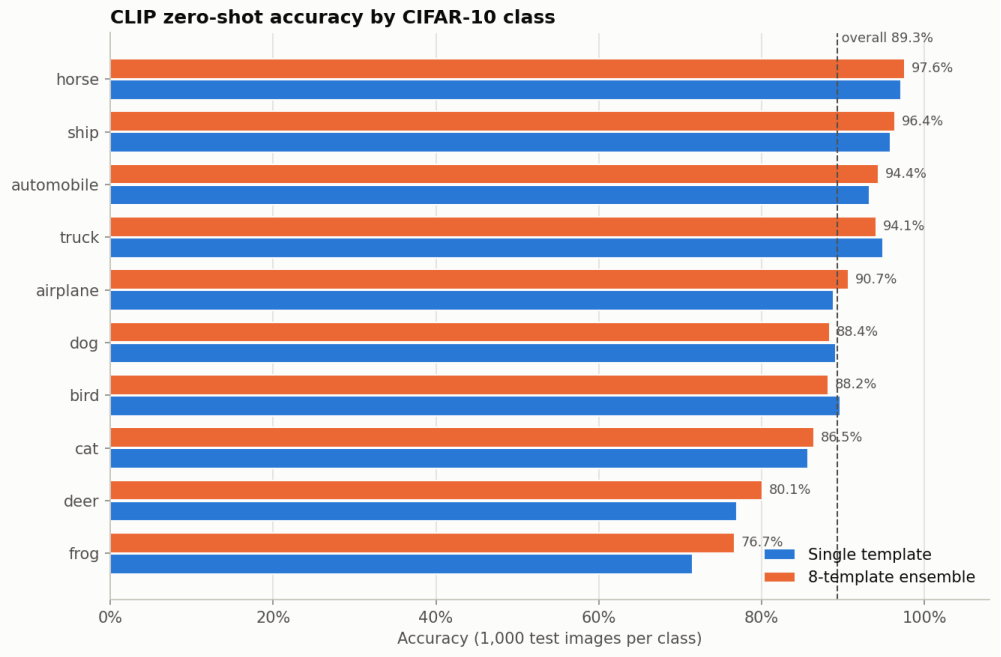


Figure 4. Per-class accuracy, single template vs. 8-template ensemble.

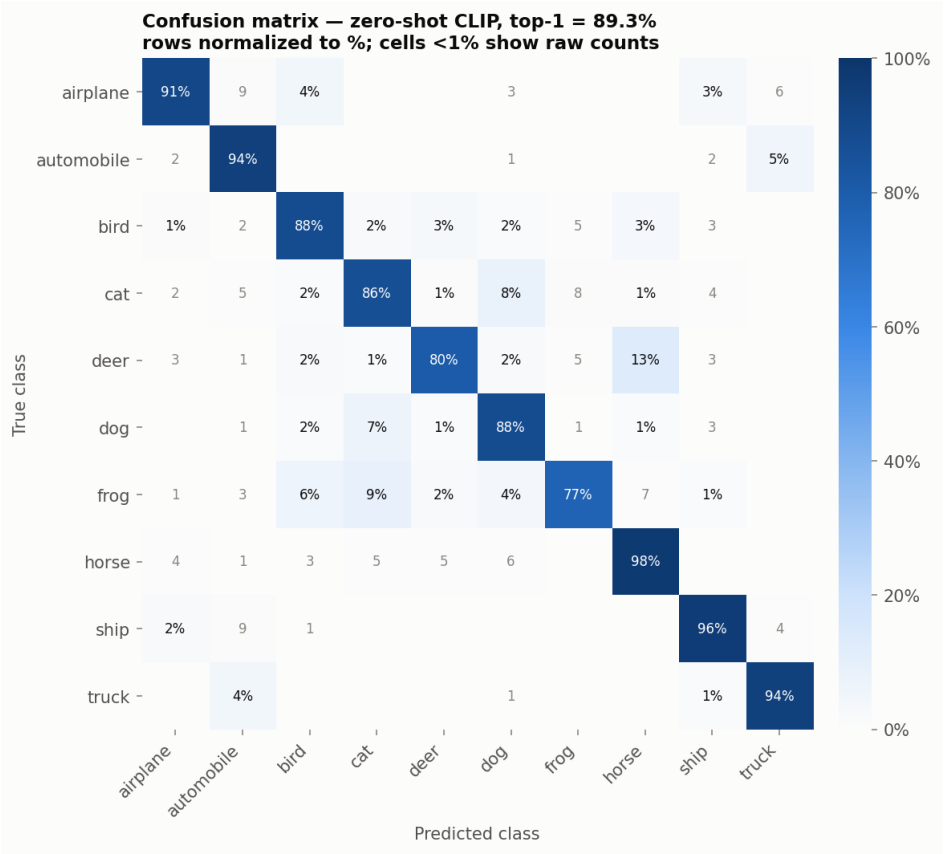


Figure 5. Confusion matrix for the ensemble (rows = true class, row-normalized; cells under 1% show raw counts).

True class	Predicted as	Images	Share of true class
deer	horse	131	13.1%
frog	cat	87	8.7%
cat	dog	81	8.1%
dog	cat	70	7.0%
frog	bird	61	6.1%
automobile	truck	51	5.1%

Table 5. The six most frequent confusions.

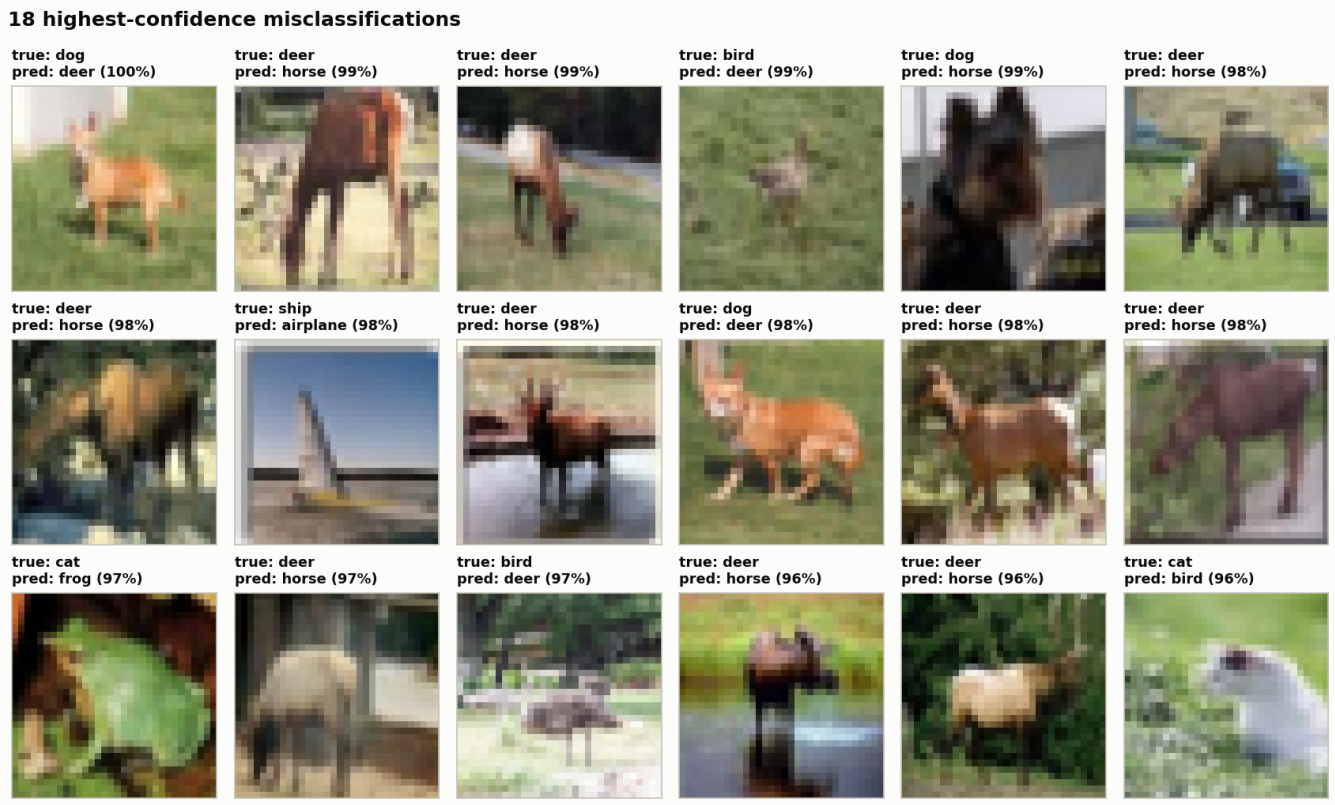


Figure 6. The 18 highest-confidence misclassifications. Several are deer→horse at 96–99% confidence; one image labeled “cat” clearly shows a frog.

Findings

1. **89.3% accuracy without training.** Matching image embeddings to class-name prompts reaches 89.31% top-1 and 97.67% top-3, showing how much transferable visual knowledge CLIP learns from natural-language supervision.
2. **Prompt ensembling helps most on the hardest classes.** The ensemble adds +1.0 pt overall and improves 7 of 10 classes, led by frog (+5.2), deer (+3.1) and airplane (+1.8). Bird (−1.5), truck (−0.8) and dog (−0.7) dip slightly. Resolution-aware prompts (“blurry”, “low resolution”) help most where the single prompt was least reliable.
3. **Errors concentrate in look-alike animals.** The largest confusion is deer→horse (131 images), followed by frog→cat, cat↔dog (81 and 70, nearly symmetric) and frog→bird. Among vehicles only automobile→truck (51) stands out; vehicles, ships and airplanes all reach 91–96%.

4 **Precision and recall are lopsided for some classes.** Horse has the highest recall (97.6%) but only 83.4% precision, because it absorbs other four-legged animals, especially deer. Frog is the reverse: 97.6% precision but 76.7% recall. At 32×32, frogs are small green or brown blobs that get spread across cat, bird and dog.

5 **Some errors are label noise or ambiguity.** A few of the most confident mistakes appear mislabeled (a “cat” that shows a frog), so part of the measured error reflects the dataset rather than the model.

4. Conclusion

CLIP's shared image–text embedding space supports both tasks with no task-specific training. A single caption retrieves its class reliably by ranking (ROC-AUC 0.963), even though absolute similarity scores are compressed and overlap between related classes. As a zero-shot classifier, CLIP ViT-B/32 reaches 89.31% on CIFAR-10, and a resolution-aware prompt ensemble adds 1.0 point. The remaining errors are mostly between visually similar animals at CIFAR-10's 32×32 resolution, plus some label noise.

Limitations and next steps. Results are for one backbone and one caption. A larger model (clip-vit-large-patch14), class-specific prompts for the confused pairs (e.g. “a photo of a deer with antlers”), or a linear probe on CLIP features would likely raise accuracy further.

Deliverables: CLIP_Similarity_ZeroShot_CIFAR10.ipynb; outputs/ (METRICS.md, metrics.json, RUN_LOG.txt, 3 CSVs, 9 figures); AI_USE.md.